AM/ME/CE 295 · Lecture 8 · Agent Design Worksheet

one page, your own project — short phrases; where a box does not apply, write why

Name: William Welch Project: Grounding-first industrial safety Team: to be confirmed

0 Research question, in one sentence
What will be true, or known, at the end that is not true now?

Does bounded adaptive grounding/semantic repair improve independently verified predicates and downstream rules over a fixed equal-budget policy?

1 Your workflow by hand
Raw input → the number you report. One step per line. Mark each step R (run it: fixed input gives fixed output) or D (look and decide: you read an output and choose what happens next).

R: load scene, allowed configuration and reviewed development references.
R: run approved visual tools; return regions, masks and candidate predicates.
D: select one bounded modification or stop/escalate.
R: execute; verify against references; log measurements and cost.
R: release accepted predicates only; evaluate fixed ILP downstream.

2 The one decision
The D step your agent makes. Input · options a / b / c · when it stops · when it escalates to a person.

Input: candidate outputs, development diagnostics, action history, budget.
Options: model/prompt/segmentation/semantic change; literature lookup; abstain.
Stop: acceptance, budget or no progress. Human: outside bounds or unresolved safety facts.

3 Its check — not the LLM
A formula, a unit test, a held-out measurement, a physical bound, a human label. Threshold from the spec or the math, never from the model.

Verifier: label agreement, box/mask overlap and binding consistency. Thresholds come from a reviewed task specification, never the agent. Held-out labels are excluded from decisions and used only for final evaluation.

4 Its baseline — same tools, no agent
A fixed rule or an optimizer on the same tools and budget. What the agent must beat; held-out cases; how many repeats.

Fixed troubleshooting order, same action catalogue/tools/start and budget.
Compare held-out predicate precision/recall/F1, false acceptance, coverage and cost; hold the ILP learner fixed.
Repeats/budget/thresholds must be frozen before testing.

5 Tool list
name(input → output, units) — one job each. Exists already? Y / N.

retrieve.py: question → chunks (exists).
check_citations.py: Grounded JSON → verdict (exists).
Visual-model wrappers: scene/config → evidence (integration planned).
Grounding verifier/gate: evidence/reference → accepted facts (planned).

6 A second agent? Why?
Only for different information, different permissions, or independent judgment. Otherwise: no — it is a tool.

One decision agent. Visual models, retriever and checker are tools, not additional decision agents. Separate human review controls labels, acceptance rules and permissions.

7 The redeploy test
Which file holds the spec · tools · permissions · memory · log · pinned model? Fresh clone, one command, smoke test passes?

Spec: CLAUDE.md / PERMISSIONS.md; skills: .claude/skills/; model/allow rules: settings.json; memory: memory/; results: eval/. Fresh-clone test, exact dependency capture and source access remain open.

8 One question for a TA
Read first tonight. The thing you could not fill in, or the thing you are least sure of.

Is the relation-annotation/acceptance protocol and equal-budget fixed troubleshooting baseline strong enough to support the grounding-first comparison?

Finish this sentence for every box you drew: **"I know this box was right because ____."** If the blank is "the model said so", the box is not done. Photograph your sheet before you hand it in. Boxes 2, 3 and 4 are the core of Sunday's proposal.